

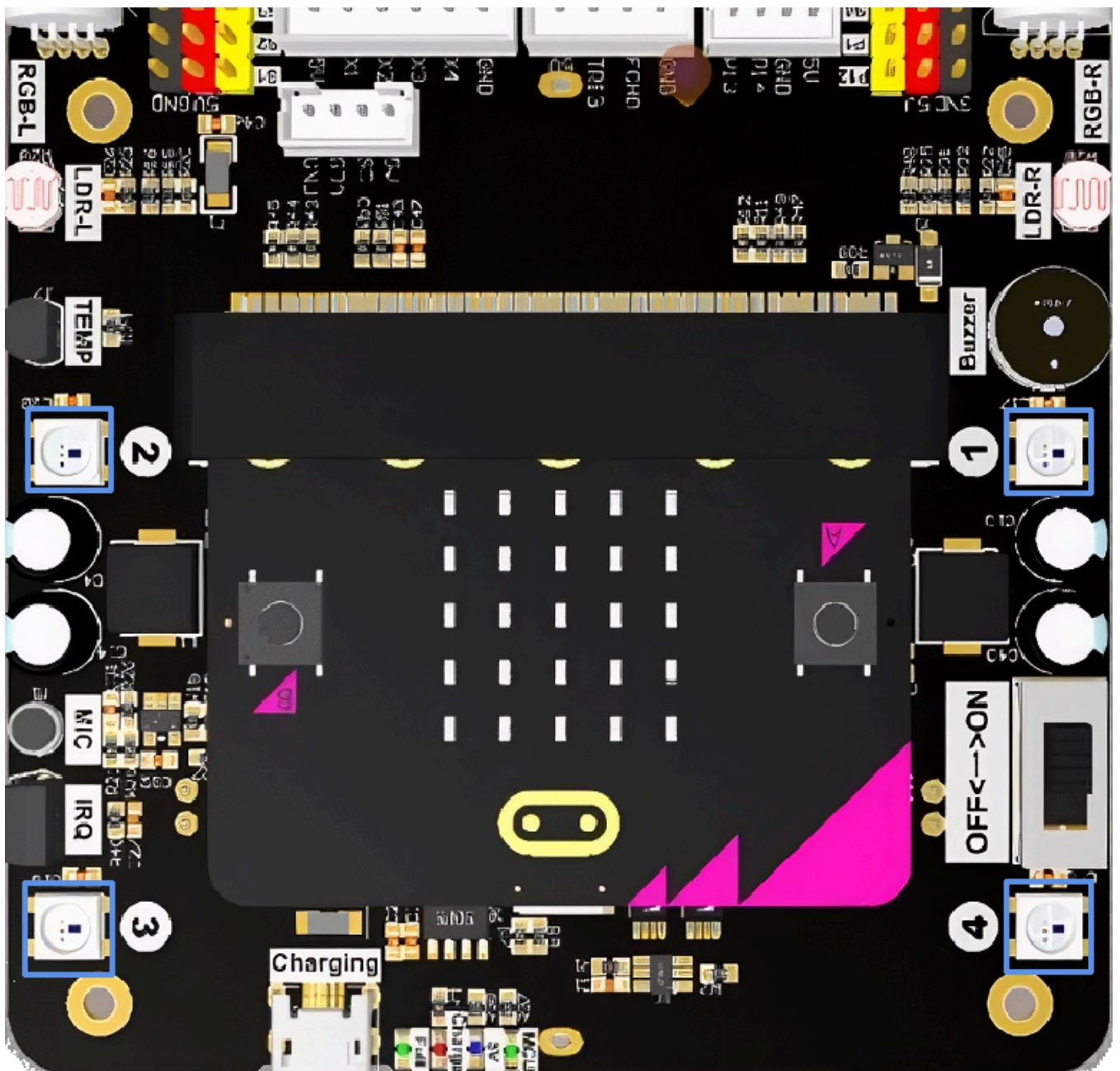
Colorful Breathing Light

Colorful Breathing Light

1. Experiment Objective
2. Experiment Steps
3. Experiment Source Code
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1. Experiment Objective

In this section, we will use building blocks to create a colorful breathing light effect.



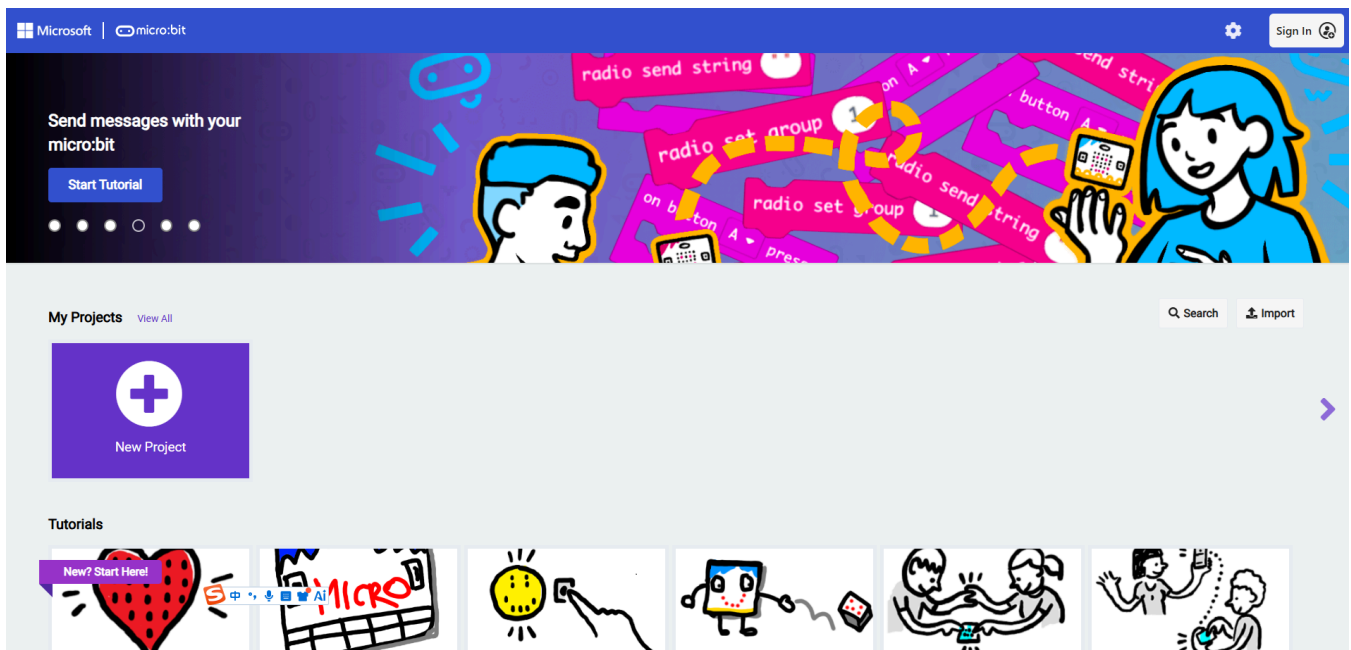
2. Experiment Steps

1. Before starting the experiment, connect the Micro:bit motherboard to your computer using a Micro USB data cable. At this time, the computer will display the MICROBIT drive letter.



2. Open MakeCode via the link below.

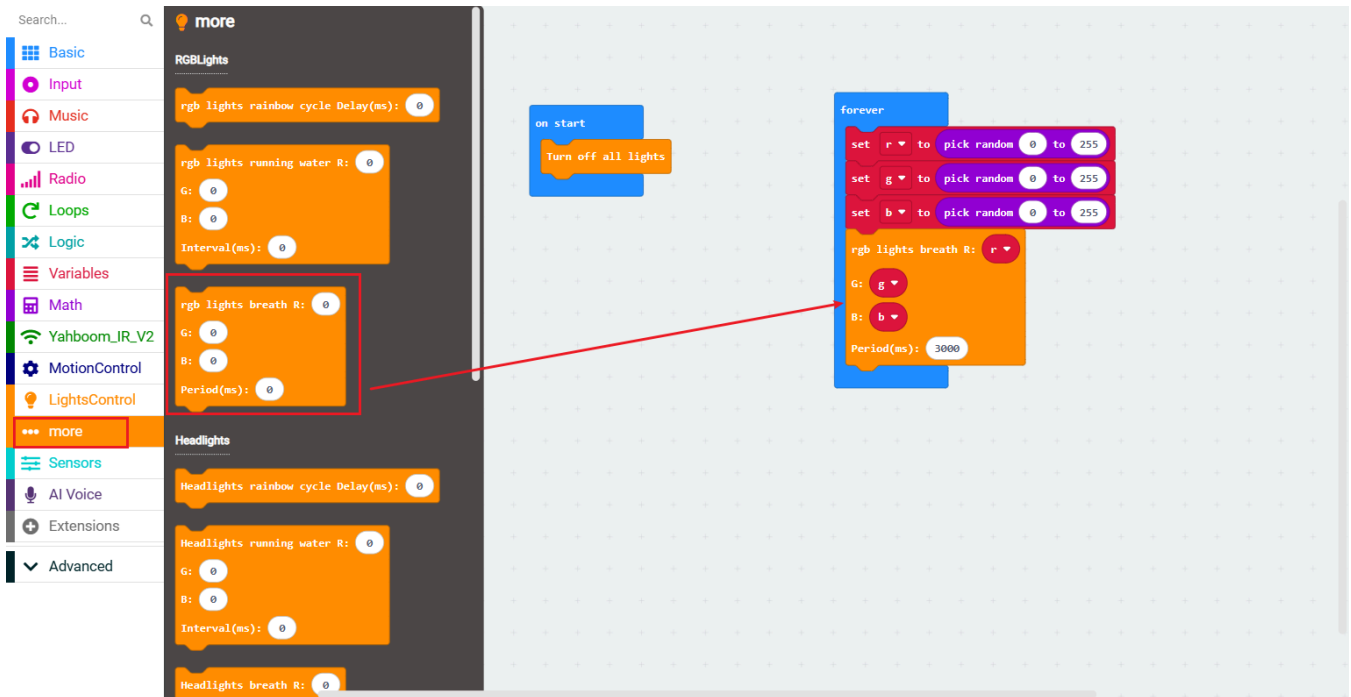
[Microsoft MakeCode for micro:bit](https://makecode.microbit.org/)



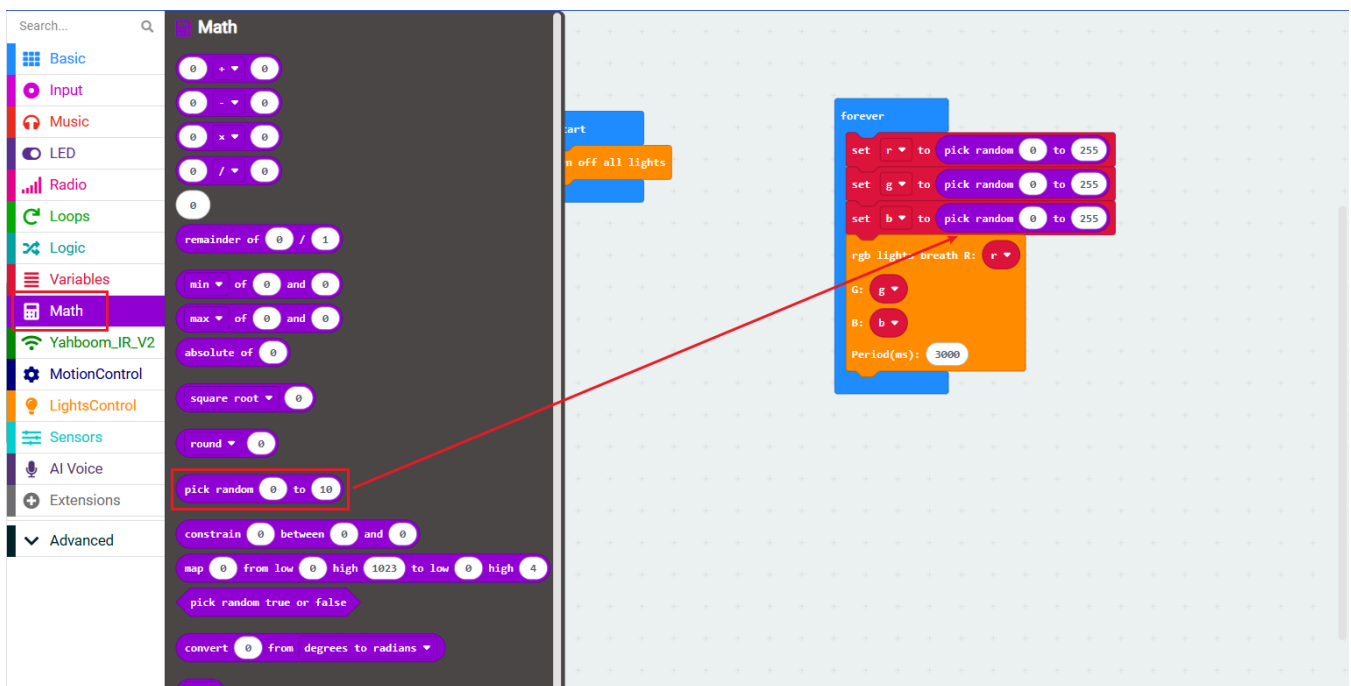
3. Drag `4.Colorful_Breathing_Light.hex` to the new project page, and MakeCode will automatically import and open the project.

3. Experiment Source Code

The building blocks for this section are in the `RGBLights` option under the `more` expansion in the `LightsControl` category.

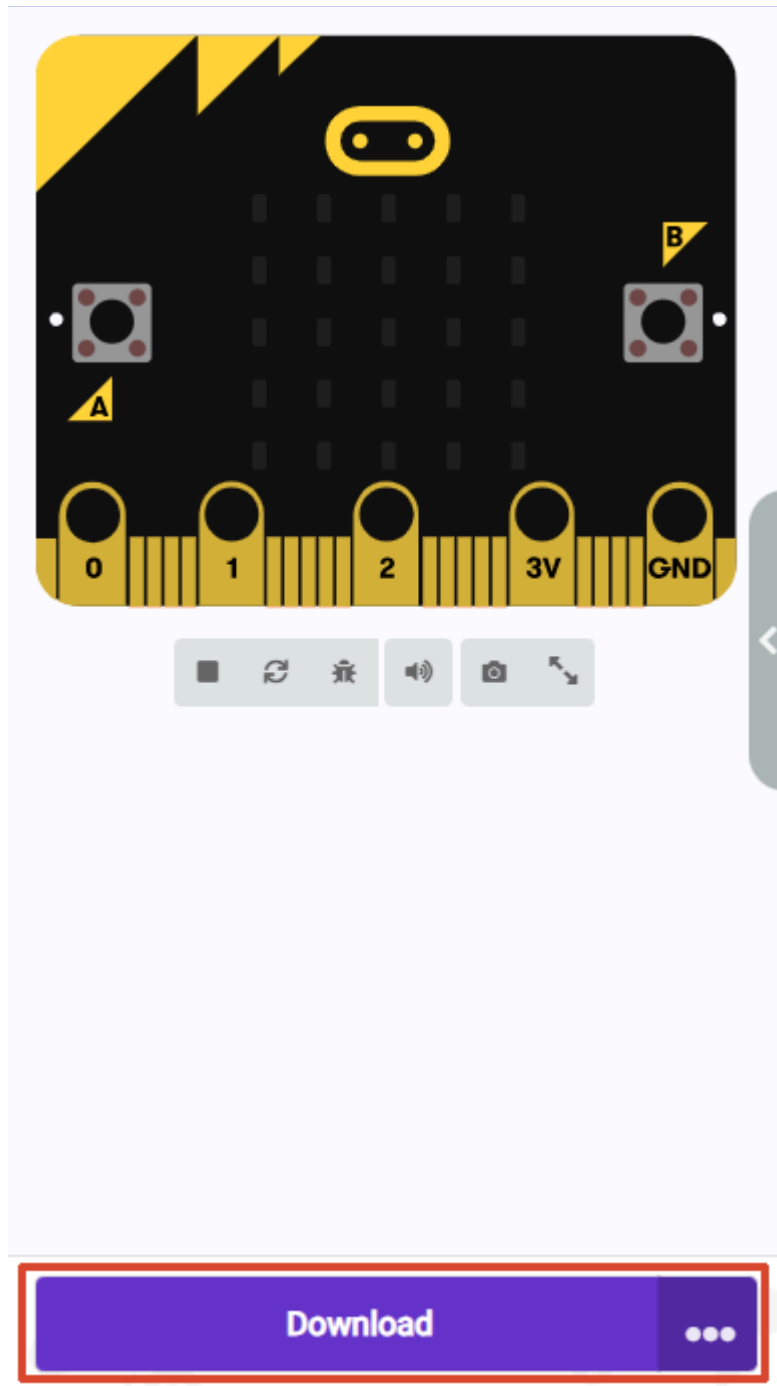


The `abs` block is located in the `Math` category.



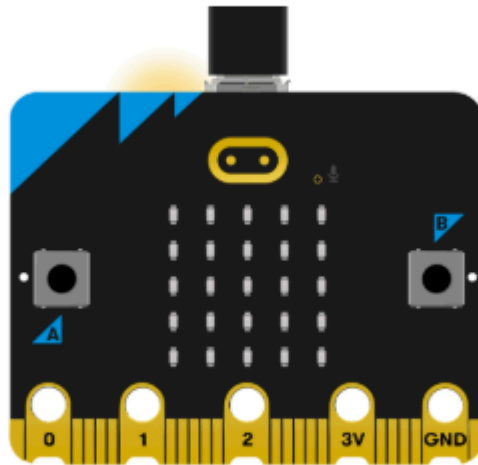
4. Program Download

1. Click the download button at the bottom left to download the program to the Micro:bit.



2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please first connect the data cable to the Microbit. If already connected, click **【Next】** here.

1. Connect your micro:bit to your computer

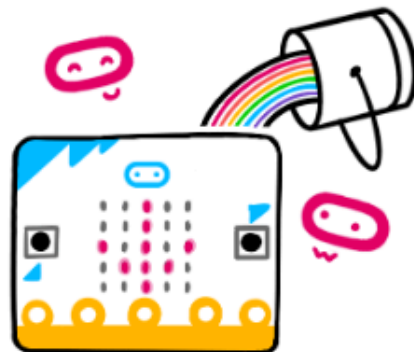


 Next

3. If the Microbit is detected to be properly connected, the following interface will appear. At this time, you can click **【Download】**.

 **Connected to micro:bit** 

Your micro:bit is connected! Pressing 'Download' will now automatically copy your code to your micro:bit.



 Download

If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. Here you can choose two methods: **【Download as File】** and **【Pair】**.

2. Pair your micro:bit to your browser



Press the Pair button below.

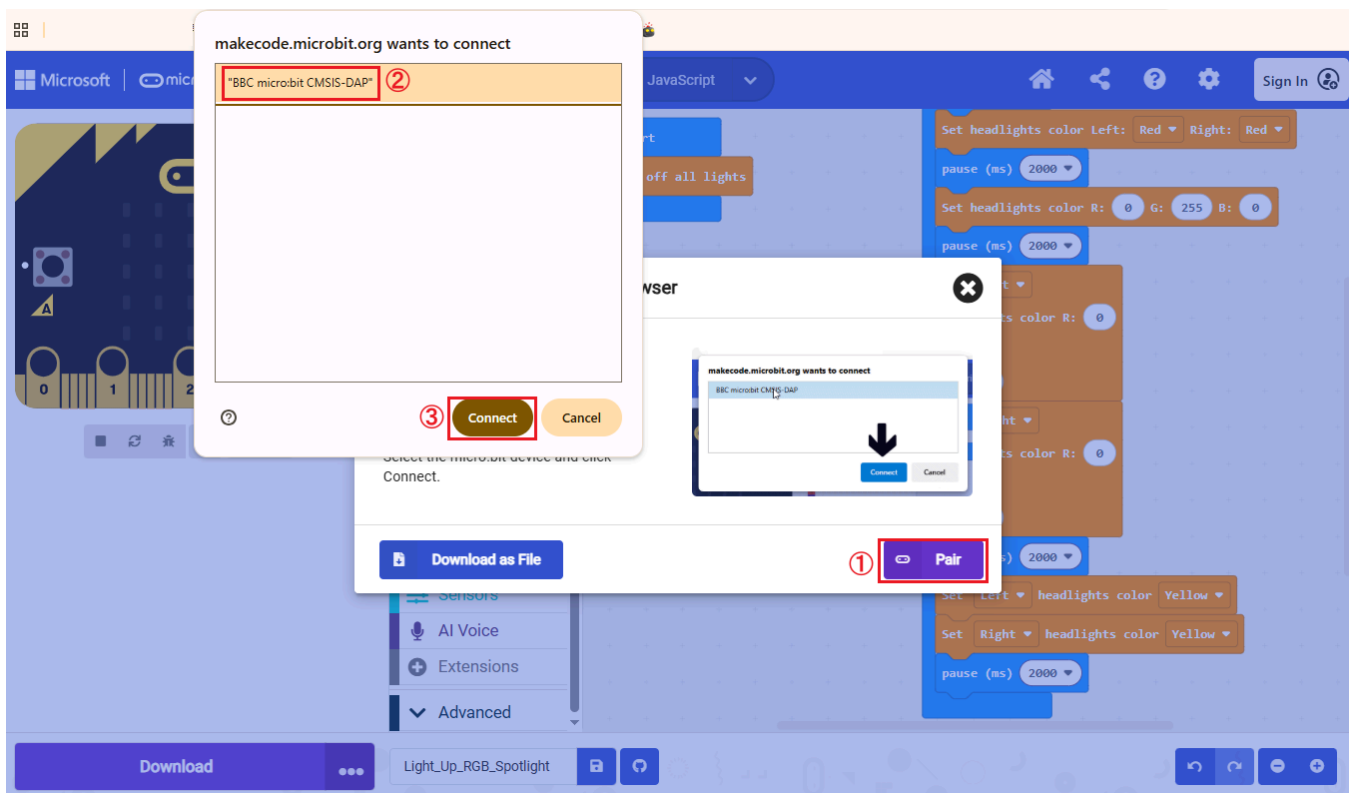
A window will appear in the top of your browser.

Select the micro:bit device and click Connect.

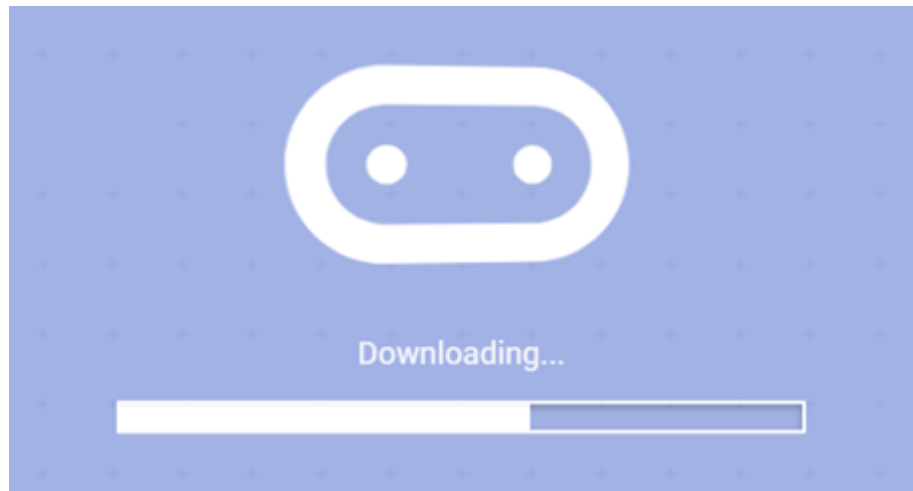


When selecting **Download as File**, the system will download the currently opened program as `microbit-project-name.hex`. You can copy or drag this program directly to the corresponding drive letter of MICROBIT to complete the download.

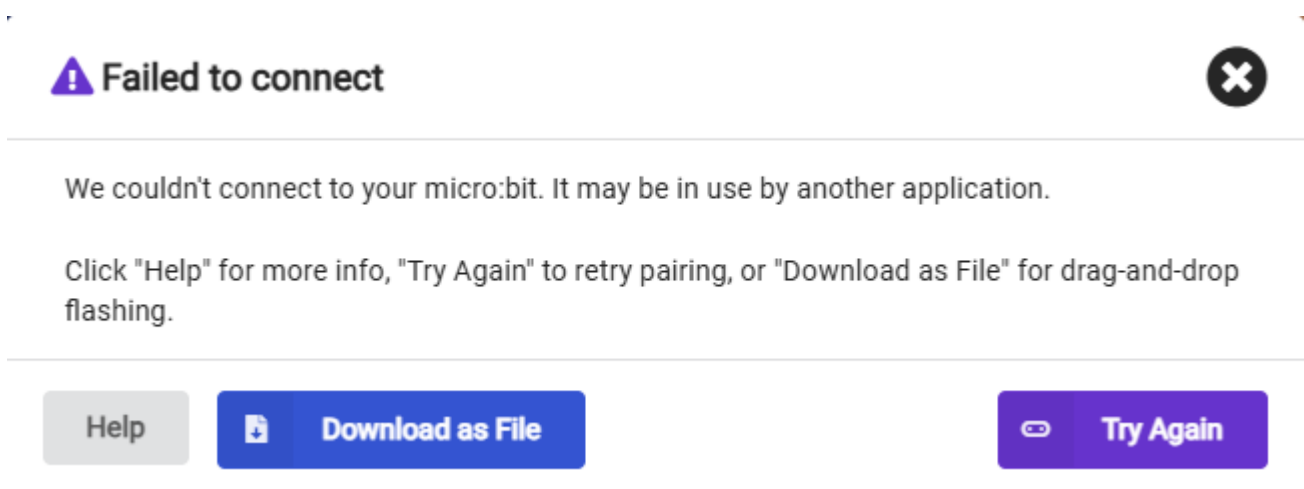
When selecting **Pair**, you will need to manually select the device. Please select `BBC micro:bit CMSIS:DAP`.



4. When the following interface appears, the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



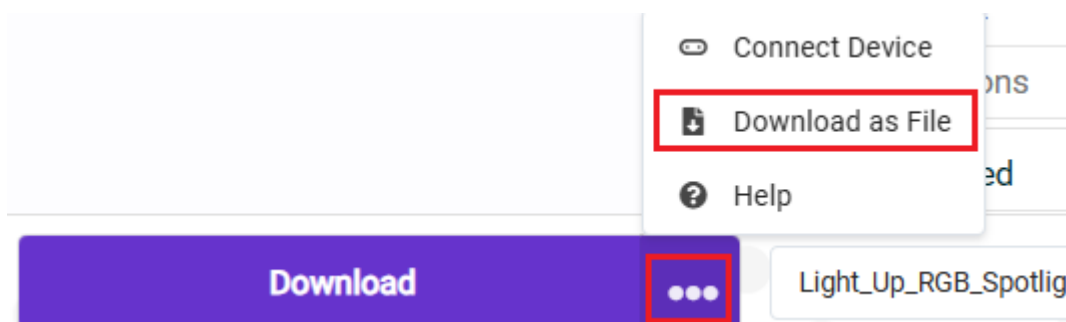
If multiple attempts always fail, directly select **【Download as File】** to save the current program as a file. Then copy or drag the program directly to the corresponding drive letter of MICROBIT to complete the download.



If you want to save the file directly, we provide two methods

Method ①:

Click **【...】** next to download, then select **【Download as File】**



Method ②:

Click **【Save】** after the project name edit field on the editing page.

5. Experiment Results

After uploading, the four colorful lights on the car will display a breathing light effect. Each effect changes every 3 seconds, and the colors will keep changing.